StayBillPro – Internship Project Dossier

Electronics Service Management & POS ERP System | S Square G Tech Solutions

Official Documentation of Intern Involvement, Project Scope & Technical Objectives

Project Overview:

StayBillPro is a specialized, production-ready ERP and billing platform built for electronics service centers, repair workshops, and multi-branch appliance service enterprises. The system centralizes the complete service lifecycle—from customer job-sheet registration and diagnosis tracking to automated GST-compliant POS invoicing, spare-parts inventory, technician allocation, and real-time business analytics.

1. INTERNSHIP PROJECT OBJECTIVES

- **Primary Project Objective:**
  To design, develop, and integrate user-centric frontend modules for StayBillPro that automate manual repair shop workflows, reduce administrative turnaround time by 60%, and deliver real-time service visibility.
- **Technical Architecture Objective:**
  Master component-driven development using React 19, modern state synchronization, dynamic client-side PDF document generation (jsPDF), barcode automation, and resilient RESTful API integration patterns.
- **ERP & Business Domain Objective:**
  Gain deep practical experience in building enterprise-grade operational systems, including tax-compliant billing (GST), inventory reordering workflows (PO & GRN), and role-based multi-tier access control.
- **Quality & Scalability Objective:**
  Implement responsive, accessible UI standards across mobile, tablet, and desktop devices, while enforcing clean code modularity, strict form validation, and edge-case testing.

2. TECHNOLOGY STACK & IMPLEMENTATION SCOPE

Technology / Library	Role in StayBillPro	Practical Skills Demonstrated
React 19 & JavaScript (ES6+)	Core UI architecture, SPA routing & dynamic components	Hooks, Component Lifecycle, Reusable UI Patterns
jsPDF & jsPDF-AutoTable	Client-side PDF generator for invoices, estimates & job sheets	Dynamic layouting, table calculations, printable exports
React Barcode & Scanner Overlay	Appliance tracking & item serial labeling	Hardware scanner integration, barcode encoding & decoding
Lucide React & CSS3 Modular	Consistent iconography, responsive layouts (Flex/Grid)	Modern UI design, responsive dashboards, dark/light styles
Razorpay API / Payment Modals	Online billing, subscription recharges & wallet management	Payment gateway webhooks, transaction status tracking
Branch & Role Security Architecture	Hierarchical permissions (Admin, Technician, Staff)	RBAC principles, route gating, sensitive data isolation

3. CORE SYSTEM MODULES DEVELOPED & MAINTAINED

Module Name	Functional Scope	Intern Involvement / Deliverables
Job Sheet & Service Ticketing	Comprehensive lifecycle tracking of faulty electronics, diagnosis reports, and technician assignments.	Designed New Job intake modal, status filters (Pending, In Progress, Delivered), and technician assignment dropdown.
POS & Wholesale Invoicing	Fast billing counter for service charges, spare parts, tax breakdowns (CGST/SGST), and discount rules.	Engineered POS cart interface, automatic total/tax calculations, and instant thermal & A4 invoice printing.
Inventory & Procurement (PO/GRN)	Stock level tracking, low-inventory alerts, purchase order creation, and goods received note processing.	Created CreatePOModal, GRN entry views, and real-time inventory deduction logic upon job completion.
Barcode Generation & Labeling	Instant barcode label generation for intake appliances to prevent unit mix-ups during multi-day repair.	Integrated react-barcode modal, formatted printable sticker sheets, and built camera-based scan overlays.
Financial Accounting & Cashbook	Daily service cashbook, technician payouts, expense logs, and GST compliance auditing tables.	Implemented ServiceCashbookScreen, ledger filter tabs, payment reconciliation screens, and data tables.
Customer & Multi-Branch Hub	Centralized CRM storing customer appliance histories, SMS/WhatsApp alerts, and branch-level filters.	Constructed customer profile view, service history timeline, and branch switching security guards.

4. KEY INTERN WORK PROFILES & RESPONSIBILITIES

Profile A: Frontend & React Developer Intern

- Engineered modular React components adhering to clean architecture and reusable design patterns.
- Integrated jsPDF and canvas libraries for dynamic invoice generation and barcode label rendering.
- Optimized client-side rendering performance, state management, and modal interaction speeds.

Profile B: UI/UX & Responsive Web Development Intern

- Designed responsive UI layouts tested on mobile service technician views and high-res cashier monitors.
- Built intuitive forms with real-time field validation, error handling, and toast feedback notifications.
- Standardized color schemes, modern card components, typography, and accessibility features.

Profile C: Quality Assurance & Integration Intern

- Formulated end-to-end user test cases spanning job creation, part allocation, and billing checkout.
- Conducted rigorous cross-browser testing (Chrome, Edge, Firefox) and responsive viewport stress testing.
- Documented edge-case bugs, API payload mismatches, and assisted in production patch deployments.

5. RESUME READY BULLET POINTS (FOR INTERN PORTFOLIOS)

Action & Impact Bullet Points (Copy & Paste for Resume / LinkedIn):

- Contributed to the frontend engineering of "StayBillPro", a multi-tenant ERP platform powering 1000+ service centers and 50k+ managed appliances.
- Architected 15+ reusable React 19 UI components for service request tickets, inventory stock logs, and technician dashboards.
- Integrated automated, client-side PDF invoice and job-sheet rendering using jsPDF and AutoTable, saving ~35 hours of manual billing per month.
- Built instant barcode sticker generation and camera-assisted scanning features, reducing device mix-ups and turnaround time by 40%.
- Collaborated in an agile team environment, conducting Git code reviews, bug fixes, and cross-browser quality assurance.

6. OFFICIAL INTERNSHIP SIGN-OFF & VERIFICATION DETAILS

Field	Details / Intern Record
Intern Name:	_____
College / Institution:	_____
Project Assigned:	StayBillPro – Service ERP & POS Platform
Organization / Company:	S Square G Tech Solutions (www.ssquareg.com)
Internship Duration:	_____ / _____ / 20___ to _____ / _____ / 20___
Assigned Role:	[ ] React Frontend Dev [ ] Full-Stack [ ] QA & Testing
Mentor / Supervisor:	Technical Lead, StayBillPro Project Team
Overall Performance:	[ ] Outstanding [ ] Excellent [ ] Commendable

Intern Signature
Date: _____

Authorized Signatory / Project Lead
S Square G Tech Solutions (Official Seal)